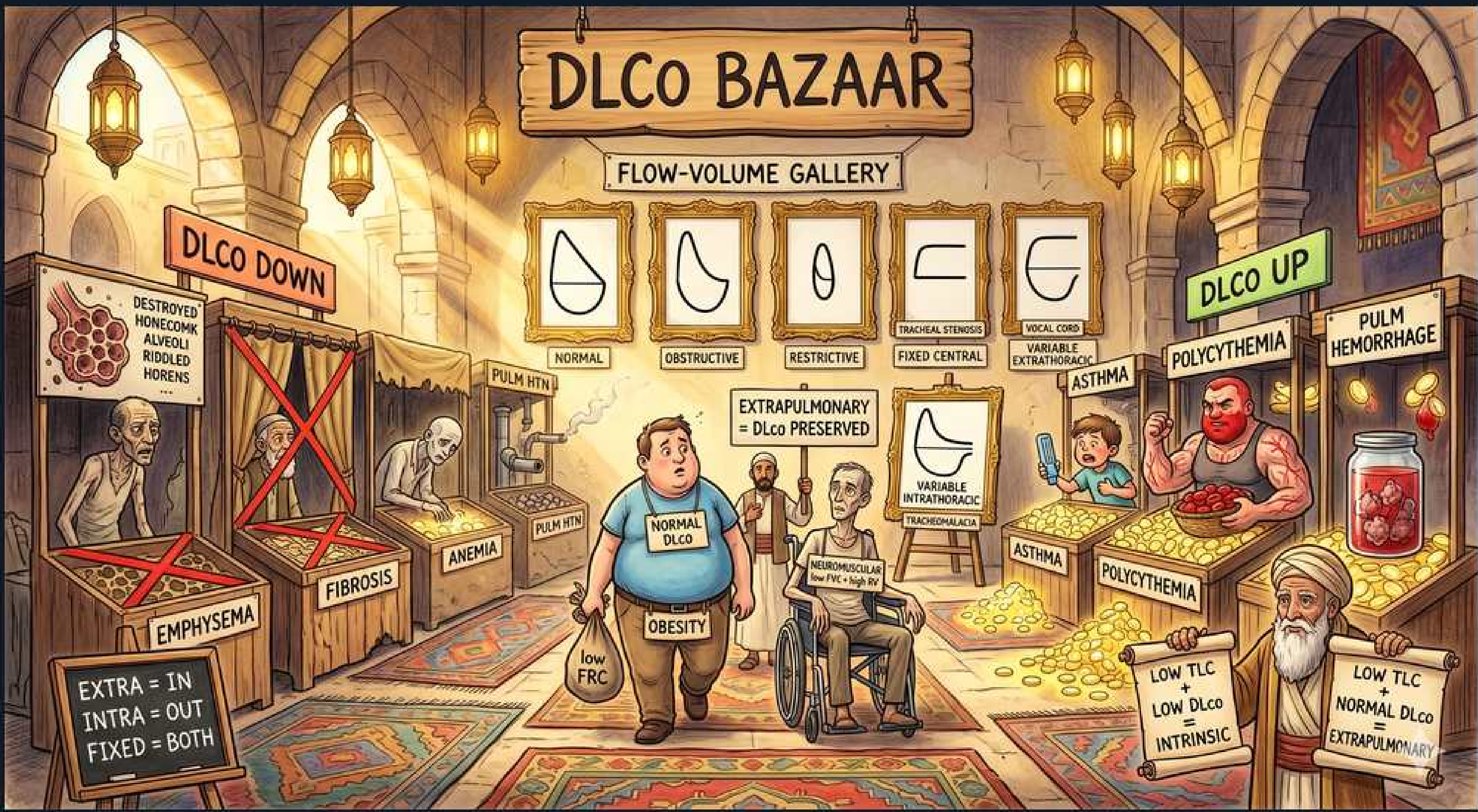
Respiratory Disturbance — DLCO Bazaar (High vs Low Causes)



1. DLCO DOWN banner

WHAT YOU SEE

Left red banner 'DLCO DOWN' over destroyed alveoli

WHAT IT ENCODES

DLCO (diffusing capacity for CO) measures how readily gas crosses the alveolar-capillary membrane and binds hemoglobin. It falls when any of three are lost: alveolar SURFACE AREA (emphysema, lobectomy), MEMBRANE integrity/thickness (ILD/fibrosis, edema), or pulmonary blood volume / HEMOGLOBIN (anemia, pulmonary vascular disease, PE/PAH). Mnemonic for low DLCO: 'emphysema, fibrosis, PE/pulmonary vascular, anemia'.

BOARD TRAP

Always correct DLCO for hemoglobin BEFORE calling it truly low — anemia alone lowers measured DLCO without any lung disease.

2. Emphysema (↓DLCO)

WHAT YOU SEE

Gaunt figure labeled EMPHYSEMA with red X

WHAT IT ENCODES

Emphysema destroys alveolar walls and capillaries → loss of surface area → low DLCO. It is the one OBSTRUCTIVE disease (FEV1/FVC <0.7) that lowers DLCO, alongside hyperinflation (↑TLC, ↑↑RV).

BOARD TRAP

Obstruction + LOW DLCO = emphysema. Obstruction with NORMAL or HIGH DLCO = asthma or chronic bronchitis. This DLCO split is the high-yield way to separate the two.

3. Fibrosis (↓DLCO)

WHAT YOU SEE

Figure labeled FIBROSIS

WHAT IT ENCODES

Pulmonary fibrosis/ILD thickens the alveolar-capillary membrane and destroys parenchyma → low DLCO with a RESTRICTIVE pattern (↓TLC, normal/high FEV1/FVC). DLCO often falls before lung volumes drop, making it an early/sensitive marker of ILD severity.

BOARD TRAP

Restriction + LOW DLCO = parenchymal lung disease (fibrosis/ILD), distinguishing it from chest-wall/neuromuscular restriction, which has a NORMAL DLCO.

4. Anemia (↓DLCO)

WHAT YOU SEE

Sign labeled ANEMIA under DLCO DOWN

WHAT IT ENCODES

DLCO depends on hemoglobin available to bind the inhaled CO. Anemia lowers measured DLCO even with perfectly normal lungs; this is corrected by the Hgb-adjusted DLCO calculation.

BOARD TRAP

A low DLCO in an anemic patient is an artifact of low Hgb — adjust for hemoglobin; it is NOT evidence of lung disease.

5. Obesity → normal DLCO

WHAT YOU SEE

Obese figure carrying bag, FRC↓, 'NORMAL DLCO'

WHAT IT ENCODES

Obesity compresses the chest wall and reduces FRC and ERV first (basal atelectasis, worse supine), giving a mildly restrictive-looking pattern. But DLCO is NORMAL or even slightly HIGH because the alveolar membrane is intact and pulmonary blood volume rises in the dependent lung.

BOARD TRAP

Obesity's restrictive-looking PFTs have a NORMAL/high DLCO — do not mistake it for fibrosis. The normal DLCO is what unmasks the extrapulmonary cause.

6. Neuromuscular → DLCO normal

WHAT YOU SEE

Wheelchair patient, 'neuromuscular = DLCO preserved'

WHAT IT ENCODES

Neuromuscular weakness and chest-wall restriction reduce the ability to inflate the lungs (↓TLC, ↓FVC) but leave the gas-exchange membrane intact → PRESERVED DLCO. (If anything, DL/VA can look high because diffusion is normalized to a small lung volume.)

BOARD TRAP

Normal DLCO + restriction = EXTRApulmonary cause (neuromuscular / chest wall / obesity), NOT ILD. A normal DLCO essentially argues against significant fibrosis.

7. Extrapulmonary = DLCO preserved

WHAT YOU SEE

Caption 'extrapulmonary = DLCO preserved'

WHAT IT ENCODES

The master rule for splitting restriction: parenchymal (lung) restriction lowers DLCO; extrapulmonary restriction (chest wall, neuromuscular, obesity, pleural disease) preserves DLCO because the lung tissue itself is normal.

BOARD TRAP

Use DLCO as the single most useful test to triage restriction: LOW = inside the lung (ILD), NORMAL = outside the lung.

8. Asthma (↑/normal DLCO)

WHAT YOU SEE

Muscular figure labeled ASTHMA on the up-side

WHAT IT ENCODES

Asthma is obstructive (reversible) but the alveoli are normal, so DLCO is NORMAL or slightly HIGH (increased pulmonary blood volume from hyperinflation pulling capillaries open). This contrasts sharply with emphysema's low DLCO.

BOARD TRAP

When both show obstruction, DLCO distinguishes asthma (normal/high) from emphysema (low) — a frequently tested discriminator.

9. DLCO UP banner

WHAT YOU SEE

Green banner 'DLCO UP' upper right

WHAT IT ENCODES

High DLCO causes more CO uptake than expected: diffuse alveolar HEMORRHAGE (intra-alveolar blood soaks up CO), POLYCYTHEMIA (extra Hgb), LEFT-TO-RIGHT shunt and early CHF (↑pulmonary blood volume), ASTHMA, EXERCISE, and the supine position. Mnemonic: 'a polycythemic asthmatic bleeding into their lungs during exercise'.

BOARD TRAP

A high DLCO is a real diagnostic clue (think alveolar hemorrhage or polycythemia) — not an artifact to dismiss.

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10. Polycythemia (↑DLCO)

WHAT YOU SEE

Figure labeled POLYCYTHEMIA holding gold coins

WHAT IT ENCODES

Polycythemia raises DLCO because excess hemoglobin binds more CO — the mirror image of anemia. This is why DLCO must be Hgb-corrected before interpretation.

BOARD TRAP

Polycythemia falsely RAISES DLCO; anemia falsely LOWERS it — opposite ends of the same hemoglobin correction.

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11. Pulmonary hemorrhage (↑DLCO)

WHAT YOU SEE

Jar of blood labeled PULM HEMORRHAGE

WHAT IT ENCODES

Diffuse alveolar hemorrhage (e.g., Goodpasture/anti-GBM, ANCA-vasculitis, SLE) fills alveoli with blood whose hemoglobin avidly binds the test CO → transiently HIGH DLCO. Serial DLCO can even track bleeding activity.

BOARD TRAP

Rising DLCO + hemoptysis + bilateral infiltrates + anemia = diffuse alveolar hemorrhage — the classic high-DLCO emergency. (Note: most other acute lung pathologies lower DLCO.)

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12. Low TLC, normal DLCO → extrapulm

WHAT YOU SEE

Bottom-right sign 'low TLC, normal DLCO, extrapulmonary'

WHAT IT ENCODES

Low TLC with a NORMAL DLCO localizes restriction OUTSIDE the lung: chest wall (kyphoscoliosis, ankylosing spondylitis), neuromuscular weakness, or obesity. The lung parenchyma and membrane are intact.

BOARD TRAP

Don't diagnose intrinsic lung disease when TLC is low but DLCO is preserved — look for an extrapulmonary cause.

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13. Low TLC, low DLCO → intrinsic

WHAT YOU SEE

Sign 'low TLC, low DLCO, intrinsic'

WHAT IT ENCODES

Low TLC WITH low DLCO localizes restriction INSIDE the lung: intrinsic parenchymal disease (ILD/fibrosis, sarcoid, hypersensitivity pneumonitis).

BOARD TRAP

Pairing TLC with DLCO is the key step that separates lung fibrosis (both low) from extrapulmonary restriction (low TLC, normal DLCO).

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14. Extra=in / Intra=out / Fixed=both

WHAT YOU SEE

Bottom-left flow-volume key 'extra=in, intra=out, fixed=both'

WHAT IT ENCODES

Upper-airway obstruction patterns on the flow-volume loop: variable EXTRAthoracic (e.g., vocal cord dysfunction, extrathoracic goiter) flattens the INspiratory limb; variable INTRAthoracic (tracheomalacia, intrathoracic tumor) flattens the EXpiratory limb; FIXED lesions (tracheal stenosis, large goiter) flatten BOTH limbs (box shape). Mechanism: extrathoracic airways collapse with the negative pressure of inspiration; intrathoracic airways collapse with the positive pleural pressure of forced expiration.

BOARD TRAP

Match the flattened limb to the lesion location — 'Extrathoracic = INspiration, Intrathoracic = EXpiration, Fixed = both.' Reversing inspiratory/expiratory is the classic board trap.
